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Department of Forensic Science, School of Sciences

Paper Presentation

Blockchain-Enabled Chain of Custody Model for Tamper-Proof Digital Evidence Management

Presented by

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OUT LINE OF THE TALK

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INTRODUCTION

- Digital evidence is important in modern investigations
- Chain of custody ensures integrity and authenticity
- Traditional systems are centralized and insecure
- Need for tamper-proof and transparent solution

LITERATURE REVIEW

- Many blockchain-based models developed (2022–2026)
- Use of Ethereum, Hyperledger, and IoT frameworks
- Smart contracts used for automation and tracking
- Research shows increasing interest in this area

METHODOLOGY

- Review of recent research papers and frameworks
- Analysis based on ISO 27037 standards
- Comparison of different blockchain models
- Focus on security, performance, and legal aspects

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RESULT

- Improved transparency and tamper detection
- Secure tracking using hashes and timestamps
- Off-chain storage improves efficiency
- Permissioned blockchain widely used

CONCLUSIONS

- Blockchain ensures secure chain of custody
- Provides tamper-proof audit trail
- Not a replacement, but supports forensic methods
- Future work needed for standardization and adoption

Challenges & Limitations

- Privacy issues with metadata exposure
- Scalability and high transaction costs
- Governance and trust management
- Legal admissibility and court acceptance
- Security of external systems (off-chain storage)

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Any Questions



Thank You